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Dynamic weighted federated contrastive learning for robust medical image classification under heterogeneous data

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Abstract

With the rapid advancement of artificial intelligence technologies, AI-driven medical image classification has attracted considerable attention in recent years. However, increasingly stringent data privacy regulations have significantly restricted data sharing among medical institutions, making it difficult for traditional centralized learning frameworks to effectively exploit distributed medical data resources. As a privacy-preserving distributed machine learning paradigm, federated learning enables multiple institutions to collaboratively train models without directly exchanging raw data. Nevertheless, federated learning still faces several critical challenges in practical applications, including heterogeneous data distributions and model bias caused by non-independent and identically distributed (Non-IID) data. To address these challenges, this paper proposes a dynamic weighted learning method based on federated contrastive learning. Different from approaches that focus solely on local representation optimization or employ fixed aggregation weights, the proposed framework couples supervised contrastive local training with loss-feedback-based multi-factor client weighting, thereby jointly controlling local representation drift and server-side model aggregation. Specifically, contrastive learning is integrated into the federated learning framework by jointly leveraging local and global model representations for supervised contrastive optimization. This strategy effectively mitigates the distribution discrepancy among different clients and gradually aligns local models with the global model as communication rounds progress. Furthermore, during global aggregation, a loss-feedback rule is introduced to accept the candidate global model parameters and aggregation weights only when the candidate loss-feedback value decreases. In addition, a multi-factor weighted aggregation mechanism is introduced to comprehensively evaluate client contributions based on multiple factors, thereby assigning larger aggregation weights to local models that are more beneficial to global model optimization. Experiments were conducted on an osteoporosis knee X-ray dataset and a COVID–Pneumonia–Normal chest X-ray dataset under IID and different Non-IID settings. Under severe Non-IID conditions, the proposed method achieved accuracies of 92.12% and 96.27%, area under the receiver operating characteristic curve (AUC) values of 98.31% and 97.68%, and F1 scores of 90.92% and 93.22% on the two datasets, respectively. Comparisons with representative federated learning methods and the static-weighting ablation

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setting demonstrate that the proposed method maintains competitive classification performance and stable training behavior under heterogeneous client distributions.

Keywords Federated learning · Supervised contrastive learning · Image classification · Non-independent and identically distributed data

Introduction

Medical image data constitute a critical component of modern medical diagnosis and treatment services, providing intuitive representations of human physiological structures and pathological alterations, thereby establishing a reliable foundation for disease diagnosis and therapeutic decision-making. As a core technology in intelligent medical image analysis, medical image classification assists physicians in rapidly and accurately interpreting multimodal medical images, thereby improving diagnostic efficiency, reducing the incidence of missed diagnoses and misdiagnoses, and providing dependable support for both clinical practice and medical research. Consequently, it plays an essential role in promoting the advancement of intelligent healthcare systems. However, traditional medical diagnosis largely depends on the individual experience and subjective judgment of physicians, which often results in limitations such as strong subjectivity, inconsistent diagnostic standards, and relatively low diagnostic efficiency.

With the rapid advancement of artificial intelligence, deep learning technologies have been extensively applied in the field of medical image processing[1–3]. Deep learning models are capable of accurately extracting and analyzing complex anatomical structures and subtle pathological features from medical images, thereby enabling physicians to diagnose diseases more efficiently and precisely and significantly promoting the development of medical image analysis. Recent studies have further demonstrated the effectiveness of deep CNN-based methods for COVID-19 classification from chest X-ray images and performance-driven lightweight CNN architectures for MRI brain tumor classification, including cross-dataset validation[4,5]. Beyond classification performance, the trustworthy application of artificial intelligence also requires systematic validation of data quality, model robustness, explainability, and governance compliance. Mukhtorov et al. developed a standards-aligned AI validation and certification platform to support reproducible and auditable model assessment[6]. Nevertheless, medical image data often contain sensitive patient privacy information and are subject to stringent legal and regulatory protections, which impose substantial barriers to cross-institutional data sharing. Meanwhile, the heterogeneity of data storage formats and management systems across different medical institutions further complicates data integration, resulting in the formation of data silos that severely constrain the training effectiveness and performance enhancement of deep learning models in medical image classification tasks.

Federated Learning (FL), a distributed machine learning paradigm, has rapidly emerged as an effective solution to the aforementioned challenges since it was first proposed by Google in 2016[7]. The fundamental principle of Federated Learning is that "data remain local while models are shared," whereby each client independently trains a local model and transmits only model parameters or gradient updates to a central server for global aggregation, thereby eliminating the need for direct raw data exchange. This framework not only provides effective privacy protection for users but also enables the efficient utilization of distributed data resources across multiple client devices and institutions.

Sheller et al.[8] reported one of the earliest applications of federated learning in multi-institutional collaboration, enabling deep learning model training without sharing patients' multimodal brain imaging data. Olivia et al.[9] demonstrated the feasibility and effectiveness of a federated learning framework using real-world electronic health records collected from one million patients. Li et al.[10] developed a practical federated learning system capable of effectively preserving privacy in brain MRI segmentation tasks. Liu et al.[11] proposed a federated learning-based framework for COVID-19 data analysis and compared the performance of four mainstream models in non-federated settings, thereby promoting subsequent research on federated learning applications for COVID-19-related

tasks[12,13]. Furthermore, Li et al.[14] employed domain-adaptive federated learning to address multi-site functional MRI image classification, while Roth et al.[15] utilized federated learning to construct a breast density classification model based on mammography imaging and reporting system data. More recently, Appasami and Savarimuthu investigated federated transfer learning for privacy-preserving MRI brain tumor classification and subsequently developed a secure federated learning framework with weighted model aggregation for heterogeneous and Non-IID multi-institutional datasets[16,17].

However, in the field of medical imaging, substantial discrepancies exist among medical institutions in terms of imaging equipment, acquisition protocols, and annotation quality, which further result in highly heterogeneous data distributions across different clients. During the iterative optimization process of federated learning, aggregating model parameters from heterogeneous clients at the server side can easily lead to update conflicts, making it difficult for the global model to converge toward the optimal solution. Moreover, repeated oscillations during the training process consume considerable computational resources and communication time, thereby adversely affecting both the convergence stability and generalization capability of the global model[18]. Although the aforementioned methods achieve satisfactory performance under weakly heterogeneous conditions, they still suffer from varying degrees of performance degradation in highly heterogeneous scenarios. In addition, most existing approaches adopt the Federated Averaging (FedAvg) algorithm[19] for model aggregation, whose fundamental strategy is to allocate aggregation weights according to the size of each client's local training dataset, such that clients with larger datasets contribute more heavily to the global model update. However, this aggregation strategy fails to fully consider the actual quality and effectiveness of local model updates, making it unreasonable to rely solely on FedAvg for model aggregation in every communication round.

To address these challenges, this paper proposes a multi-factor dynamic weighting method that integrates federated learning with supervised contrastive learning. In the proposed framework, each client updates its local model using its private data, while supervised contrastive learning is employed to constrain the representation discrepancy between the local model and the global model, thereby guiding local models toward more consistent and discriminative feature representations. During global aggregation, both the local training loss and the local dataset size are incorporated into the calculation of client aggregation weights, allowing the server to evaluate client contributions using multiple factors. The candidate global model parameters and aggregation weights are conditionally accepted according to changes in the loss-feedback value between consecutive communication rounds. When the candidate loss-feedback value decreases, the newly calculated global model parameters and aggregation weights are accepted; otherwise, the global model parameters and aggregation weights from the previous round are retained to reduce fluctuations caused by unstable local updates.

By integrating local representation alignment and server-side contribution adjustment into a unified federated training process, the proposed method aims to alleviate client drift and aggregation instability under Non-IID data distributions. Experimental results on two medical X-ray image datasets demonstrate that the proposed method achieves stable classification performance under different levels of data heterogeneity. The main contributions of this paper are summarized as follows:

- A federated supervised contrastive learning framework is developed to jointly coordinate local representation learning and global model aggregation. The supervised contrastive constraint guides local models toward more consistent and discriminative feature representations under heterogeneous client data distributions.
- A multi-factor aggregation strategy is proposed by jointly considering client training loss and local dataset size, enabling the server to evaluate client contributions based on both local training performance and data scale.
- A loss-feedback-based update mechanism is introduced to determine whether the candidate global model parameters and aggregation weights are accepted according to changes in the loss-feedback value across communication rounds. Its effectiveness is evaluated on two medical X-ray datasets through comparisons with representative federated learning methods and a targeted static-weighting ablation study.

Related work

Federated learning in image classification

Federated learning enables efficient model training without transferring raw data from local devices by coordinating machine learning optimization across multiple distributed devices or institutions[20], thereby effectively utilizing decentralized data resources while preserving data privacy. McMahan et al.[7] proposed the FedAvg algorithm, which generates a globally updated model by computing the weighted average of model parameters uploaded from individual clients. Due to its simplicity and practicality, FedAvg has become one of the most widely adopted aggregation algorithms in federated learning. However, under Non-IID data distributions, FedAvg often suffers from issues such as unstable convergence, significant performance fluctuations, and even training failure. To address these limitations, numerous studies have proposed optimized aggregation strategies for different heterogeneous data scenarios. Building upon FedAvg, Li et al.[21] proposed the FedProx algorithm, which introduces a regularization term to constrain the deviation between local model updates and global model parameters within a certain range. Although this strategy alleviates the impact of data heterogeneity to some extent, it also restricts the flexibility of local model optimization. Wang et al.[22] proposed the FedMA algorithm for multi-task federated learning scenarios. FedMA is a federated matched averaging method specifically designed for modern neural network architectures, such as Convolutional Neural Networks (CNNs) and Long Short-Term Memory (LSTM) networks. The algorithm performs feature matching and weighted aggregation on hidden-layer neurons, thereby constructing a more consistent and efficient global model across multiple clients. Furthermore, the Per-FedAvg algorithm[23] extends FedAvg by incorporating personalized optimization strategies, enabling each client to conduct model training tailored to its own local data distribution. This method demonstrates superior performance in highly heterogeneous federated learning environments. Li et al. proposed Model-Contrastive Federated Learning (MOON), which employs model-level contrastive learning to constrain local training by encouraging consistency between the representations of the current local model and the global model. This strategy reduces local model drift caused by heterogeneous client data and improves federated learning performance under Non-IID distributions[24].

Federated learning has attracted increasing attention in medical imaging applications owing to its unique capability to preserve data privacy while enabling collaborative model training[25–30]. Ogier et al.[31] proposed a predictive diagnostic framework for triple-negative breast cancer by integrating machine learning within a federated learning architecture, where patient data remained securely stored behind hospital firewalls throughout the training process. Experimental results demonstrated that federated training further improved the predictive performance for evaluating patients' histological response to neoadjuvant chemotherapy. Roth et al.[15] applied federated learning methods to construct a practical medical image classification model, and their findings showed that the FL-trained model outperformed models trained exclusively on local datasets by an average of 6.3%. This result further verified the effectiveness of federated learning in real-world scenarios with limited datasets and computational resources. In addition, Luo et al.[32] proposed a medical image classification framework named FedSLD to address the problem of heterogeneous data distributions across multiple clients. In FedSLD, the contribution of each data sample to the local optimization objective is adaptively adjusted according to the label distribution of individual clients, thereby alleviating the training instability caused by data heterogeneity.

Contrastive learning in image classification

In recent years, contrastive learning[33–35] has emerged as a prominent research direction in self-supervised learning[36–38] owing to its ability to learn discriminative feature representations from unlabeled data. The fundamental idea of contrastive learning is to perform representation optimization in the feature space by pulling feature embeddings of positive sample pairs closer together while pushing negative sample pairs farther apart, thereby enabling the model to learn more distinguishable data representations. Supervised contrastive learning further

extends traditional contrastive learning by incorporating label information[39] to explicitly define positive and negative sample relationships. Compared with conventional self-supervised contrastive learning methods, supervised contrastive learning enables the model to more accurately capture inter-class separability and intra-class consistency, thereby effectively improving representation learning efficiency and overall model performance.

Contrastive learning has demonstrated remarkable effectiveness in leveraging unlabeled data, addressing few-shot learning problems, and improving model interpretability. Owing to these advantages, it has become a highly influential and widely adopted research direction in the field of medical imaging. Zhang et al.[40] employed a dual-branch contrastive framework to optimize the consistency between generated views and the original graph structure, thereby improving graph data classification performance. Yang et al.[41] proposed the SimCLIC framework, which simplifies the conventional dual-network architecture by eliminating the dependence on negative sample pairs and momentum encoders, and achieves contrastive learning through perturbations of image representations, demonstrating competitive performance across multiple benchmark image classification datasets. Furthermore, Zhang et al.[42] introduced the TA-DCL method, which integrates triple-attention mechanisms with dual-pooling contrastive learning for multi-label medical image classification, achieving superior performance compared with existing state-of-the-art approaches on the ODIR and NIH-ChestXray14 datasets. Kuang et al.[43] proposed PiCCL, a lightweight multi-view contrastive learning framework that adopts a multi-branch network architecture and improves learning efficiency through a customized loss function. Experimental results showed that PiCCL achieved excellent image classification performance on datasets such as CIFAR-10 and STL-10, particularly in few-shot learning scenarios. Overall, contrastive learning has been extensively explored and successfully applied in both supervised and unsupervised learning paradigms.

Existing studies have improved federated learning under heterogeneous data distributions mainly from the perspectives of local optimization, representation learning, or server-side aggregation. In this study, supervised contrastive learning is incorporated into local training to constrain the representation discrepancy between local and global models, while client training loss and local dataset size are jointly considered during global aggregation. The aggregation weights are further adjusted according to changes in the server-side loss-feedback value across communication rounds. By combining representation alignment with loss-feedback-based client weighting, the proposed framework coordinates local model optimization and global model aggregation within a unified training process.

Method

In this section, we first present an overview of federated learning, followed by a detailed introduction to the framework and network architecture of federated supervised contrastive learning. Subsequently, the optimization objective and the hybrid training loss function adopted in this study are described in detail. Finally, the proposed multi-factor dynamic weighting method is comprehensively described, along with the complete training algorithm presented in this paper.

Definition of federated learning

Suppose there is one server S and K clients $C = \{C_1, C_2, \ldots, C_K\}$, each of which possesses its own local dataset $D = \{D_1, D_2, \ldots, D_K\}$. In the context of federated learning, we need to train effective models without directly sharing data; therefore, a central server is used to aggregate the parameters of the local models. The federated learning framework is shown in Fig. 1, and the standard training process is as follows:

(1) Initialize the model on the central server and distribute the global model parameters w_G to the clients.
(2) Each client C_k trains local model parameters $w_k\,(k = 1, 2, \ldots, K)$ on its own dataset D_k.
(3) Client C_k sends the local model parameters w_k to the central server.

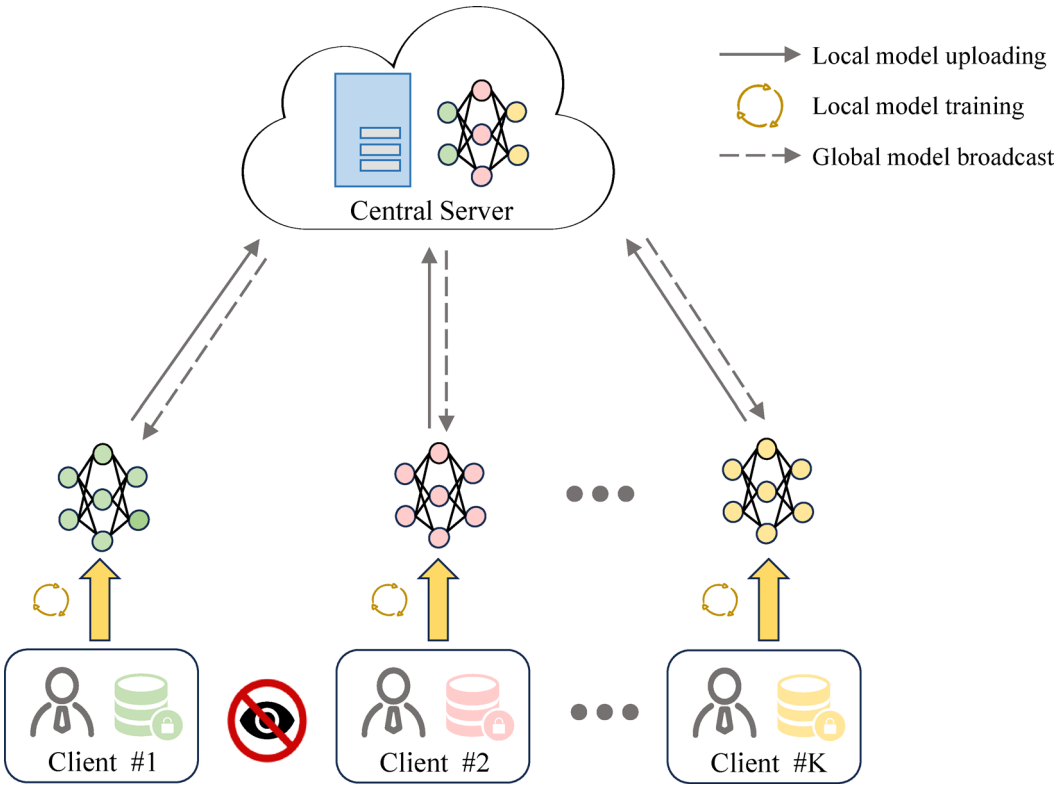


Fig. 1 Federated learning framework.

(4) The central server aggregates the local model parameters from each client and updates the global model parameter $w_G \leftarrow \sum_{k=1}^{K} \frac{N_k}{N} w_k$, where $N = \sum_{k=1}^{K} N_k$ is the total number of data points in the federated learning setup.

During the federated learning process, the above steps are iteratively executed over multiple communication rounds until the global model converges and achieves the expected performance. Throughout the entire training procedure, all raw training data remain strictly stored on local devices, while information exchange between the central server and participating clients is limited to model parameters or gradient updates. Consequently, federated learning enables collaborative model training without direct data sharing, thereby effectively alleviating the problem of data silos.

Federated supervised contrastive learning framework

The fundamental principle of federated supervised contrastive learning is to minimize the distance between the feature representations of positive sample pairs while maximizing the distance between negative sample pairs, thereby enabling the model to learn more discriminative feature representations. Specifically, during each communication round, the global model serves as a shared reference for guiding representation learning across local clients. The key objective is to ensure that the feature representations generated by local models across different clients remain aligned with those of the global model from the previous communication round, while simultaneously preserving sufficient distinction from the feature representations produced by the local models in the previous iteration.

In medical image classification tasks, this framework is used to reduce the representation discrepancy caused by heterogeneous client data distributions. The overall architecture is illustrated in Fig. 2. During each communication round, the server distributes the current global model to the participating clients, and each client updates

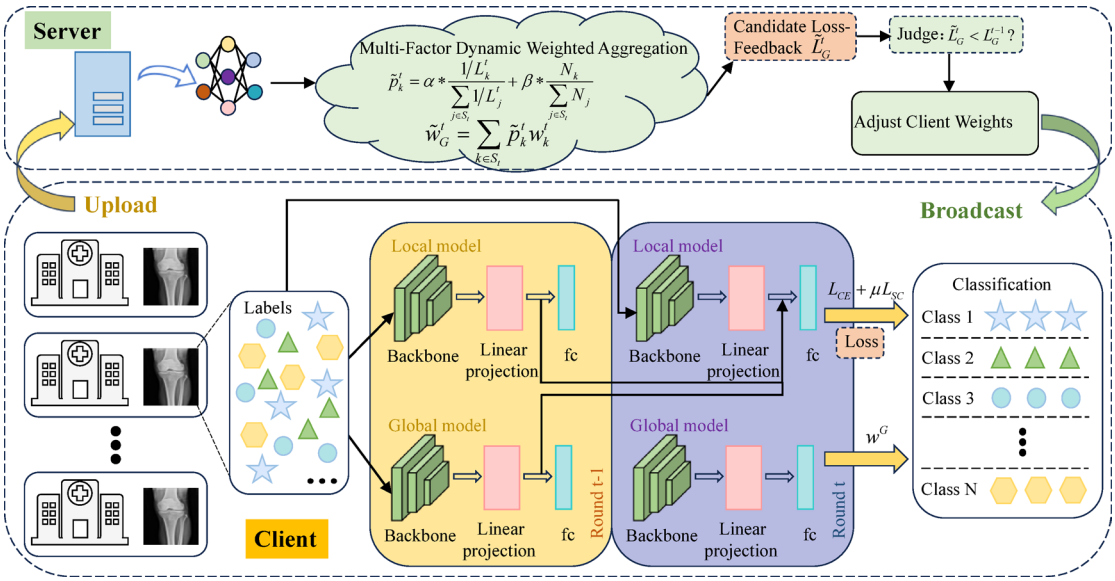


Fig. 2 Overall framework diagram.

a local model using its private medical image data. The backbone network extracts image features, which are subsequently mapped into the contrastive feature space through a linear projection layer. In parallel, a fully connected classification head generates the class predictions.

During local training, the model optimization process is jointly driven by classification supervision and contrastive constraints. Specifically, cross-entropy loss is employed to ensure classification performance, while positive and negative sample pairs are constructed based on label information. Meanwhile, supervised contrastive loss is introduced to constrain the feature distribution by reducing the distance between positive sample pairs and enlarging the distance between negative sample pairs within the feature space. Furthermore, a dual-branch architecture consisting of the current local model and the global model from the previous communication round is constructed to achieve representation alignment in a shared feature space, thereby ensuring that local model updates are guided by global knowledge constraints. In the t-th communication round, the local model is jointly optimized with the global model from the (t−1)-th round under a hybrid loss constraint, thereby enhancing the consistency and discriminative capability of the learned feature representations.

After local training is completed, each client uploads its model parameters along with the corresponding loss information to the central server. Subsequently, the server performs global aggregation using the proposed multi-factor dynamic weighting strategy, which comprehensively considers both the local training loss and the scale of the local dataset, thereby generating an updated global model for redistribution to all participating clients. Through multiple rounds of iterative optimization, the overall model performance is progressively improved until convergence is achieved. The hybrid training loss function and the proposed dynamic weighting aggregation algorithm employed in this framework will be described in detail in the following sections.

Optimization objective

During the local training phase of client C_k in the t-th communication round, the current local model is w_k^t and the global model is w_G^t. Let F_w^x denote the features obtained by passing the local data sample x through the linear projection layer of the network model. It follows that the input data sample x can be represented as $V_G^{t-1}\left(V_G^{t-1}=F_{\omega_G^{t-1}}^x\right)$ using features extracted by the previous-round global model w_G^{t-1}, as $V_k^{t-1}\left(V_k^{t-1}=F_{\omega_k^{t-1}}^x\right)$ using features extracted by the previous-round local model w_k^{t-1}, and as $V_k^t\left(V_k^t=F_{\omega_k^t}^x\right)$ using features extracted by the current-round local model

w_k^t. Based on the above definitions, the objective of local model training is to minimize the distance between V_k^t and V_G^{t-1} while maximizing the distance between V_k^t and V_k^{t-1}. Similar to the NT-Xent loss function[44], our definition of the supervised contrastive loss function is shown in Eq. (1):

$$L_{SC} = -\log \frac{\exp\left(sim\left(V_k^t, V_G^{t-1}\right)/\tau\right)}{\exp\left(sim\left(V_k^t, V_G^{t-1}\right)/\tau\right) + \exp\left(sim\left(V_k^t, V_k^{t-1}\right)/\tau\right)} \tag{1}$$

Here, $sim(\cdot)$ represents the cosine similarity function between features, and τ is an adjustable temperature coefficient.

This paper employs the cosine similarity function as a metric to construct a supervised contrastive loss. This function reflects the similarity between two features in a space by characterizing the angle between them; the smaller the angle, the more similar the two features are. For two features $X = (x_1, x_2, \ldots, x_n)$ and $Y = (y_1, y_2, \ldots, y_n)$ in an n-dimensional space, the cosine similarity function is given by Eq. (2):

$$\cos(X, Y) = \frac{\sum_{i=1}^{n}(x_i \cdot y_i)}{\sqrt{\sum_{i=1}^{n} x_i^2} \cdot \sqrt{\sum_{i=1}^{n} y_i^2}} \tag{2}$$

Therefore, our training loss function can be expressed by Eq. (3):

$$L = L_{CE}\left(w_k^t, x\right) + \mu L_{SC}\left(w_k^t, w_k^{t-1}, w_G^{t-1}, x\right) \tag{3}$$

In Eq. (3), μ is a hyperparameter that controls the contribution of the supervised contrastive loss.

For the proposed federated supervised contrastive learning method, the optimization objective in this paper is composed of the cross-entropy loss function L_{CE} and the supervised contrastive loss function L_{SC}. Consequently, the optimization objective of the global model can be expressed by Eq. (4):

$$\arg\min_{w_G} L(w_G) = \sum_{k=1}^{K} \frac{n_k}{n} L_k(w_G) \tag{4}$$

In Eq. (4), $L_k(\cdot)$ represents the training loss function of the client C_k.

Loss-feedback-based multi-factor dynamic weighted aggregation

During the federated aggregation phase, traditional federated learning methods typically rely solely on local dataset size for aggregation weighting, making it difficult to accurately reflect the differences in update quality among clients under Non-IID data distributions. As a result, low-quality local updates may be introduced into the global aggregation process, thereby adversely affecting the performance of the global model. To address this issue, this paper incorporates both the client training loss and the size of the local training dataset into the aggregation weight calculation, thereby enhancing the convergence stability and generalization capability of the global model. The basic formulation is presented in Eq. (5):

$$\tilde{p}_k^t = \alpha * \frac{1/L_k^t}{\sum_{j \in S_t} 1/L_j^t} + \beta * \frac{N_k}{\sum_{j \in S_t} N_j} \tag{5}$$

Here, L_k, and N_k denote the local training loss, the number of local training samples, and the local model parameters of client k in communication round t, respectively. The coefficients α and β control the contributions of the loss-based term and the sample-size-based term, respectively, and satisfy $\alpha + \beta = 1$. The set of participating clients in communication round t is denoted by S_t.

For clarity, the aggregation weight calculated using Eq. (5) is denoted as the candidate aggregation weight $\tilde{p}_k^t$ for client k in the t-th communication round. The candidate global model parameters are obtained as

$$\tilde{w}_G^t = \sum_{k \in S_t} \tilde{p}_k^t w_k^t \tag{6}$$

w_k^t denotes the local model parameters uploaded by client k in communication round t.

The final global model and aggregation weights are determined according to the loss-feedback rule:

$$(w_G^t, p^t, L_G^t) = \begin{cases} (\tilde{w}_G^t, \tilde{p}^t, \tilde{L}_G^t), \tilde{L}_G^t < L_G^{t-1} \\ (w_G^{t-1}, p^{t-1}, L_G^{t-1}), \tilde{L}_G^t \geq L_G^{t-1} \end{cases} \tag{7}$$

Here, $\tilde{L}_G^t$ denotes the sample-size-weighted aggregate of the local training losses reported by the participating clients in communication round t, and it is used as the server-side loss-feedback value for evaluating the candidate aggregation result. L_G^{t-1} denotes the previously accepted loss-feedback value. These quantities are calculated from client-reported loss information and do not require the server to access raw medical data.

Building on the above formulation, the dynamic aggregation process is illustrated in Fig. 3. In each communication round, the server receives the local model parameters and loss information uploaded by the participating clients, calculates the candidate aggregation weights according to Eq. (5), and obtains the candidate global model parameters according to Eq. (6). The server then determines whether the candidate global model parameters and aggregation weights should be accepted according to the loss-feedback rule defined in Eq. (7). The accepted global model parameters are subsequently broadcast to the participating clients for the next communication round.

Stability analysis of the dynamic aggregation rule

Assuming that the local training loss of each participating client is positive, the loss-based term and the dataset-size-based term in Eq. (5) are both non-negative and separately normalized over the participating client set S_t. Because $\alpha \geq 0, \beta \geq 0$, and $\alpha + \beta = 1$, the candidate aggregation weights satisfy:

$$\tilde{p}_k^t > 0, \sum_{k \in S_t} \tilde{p}_k^t = 1 \tag{8}$$

Therefore, according to Eq. (6), the candidate global model parameters are obtained by taking a normalized weighted average of the local models uploaded by the participating clients. Since all aggregation weights are non-negative and sum to one, the aggregation process preserves a valid and bounded weighting structure.

According to the loss-feedback rule in Eq. (7), the candidate global model parameters and aggregation weights are accepted only when the candidate loss-feedback value is lower than the previously accepted loss-feedback value. Therefore, the accepted loss-feedback value satisfies:

$$L_G^t = \min \{\tilde{L}_G^t, L_G^{t-1}\} \tag{9}$$

and consequently,

$$L_G^t \leq L_G^{t-1} \tag{10}$$

Because the loss function is non-negative, the sequence of accepted loss-feedback values is monotonically non-increasing and bounded from below, and therefore converges to a finite limit. When the candidate update is rejected, the global model and aggregation weights remain unchanged:

$$w_G^t = w_G^{t-1}, p^t = p^{t-1} \tag{11}$$

Thus, when a candidate update is rejected, neither the global model nor the aggregation weights are changed, thereby avoiding additional fluctuations in that communication round. These properties provide theoretical support for the monotonic acceptance behavior and aggregation stability of the proposed dynamic weighting mechanism,

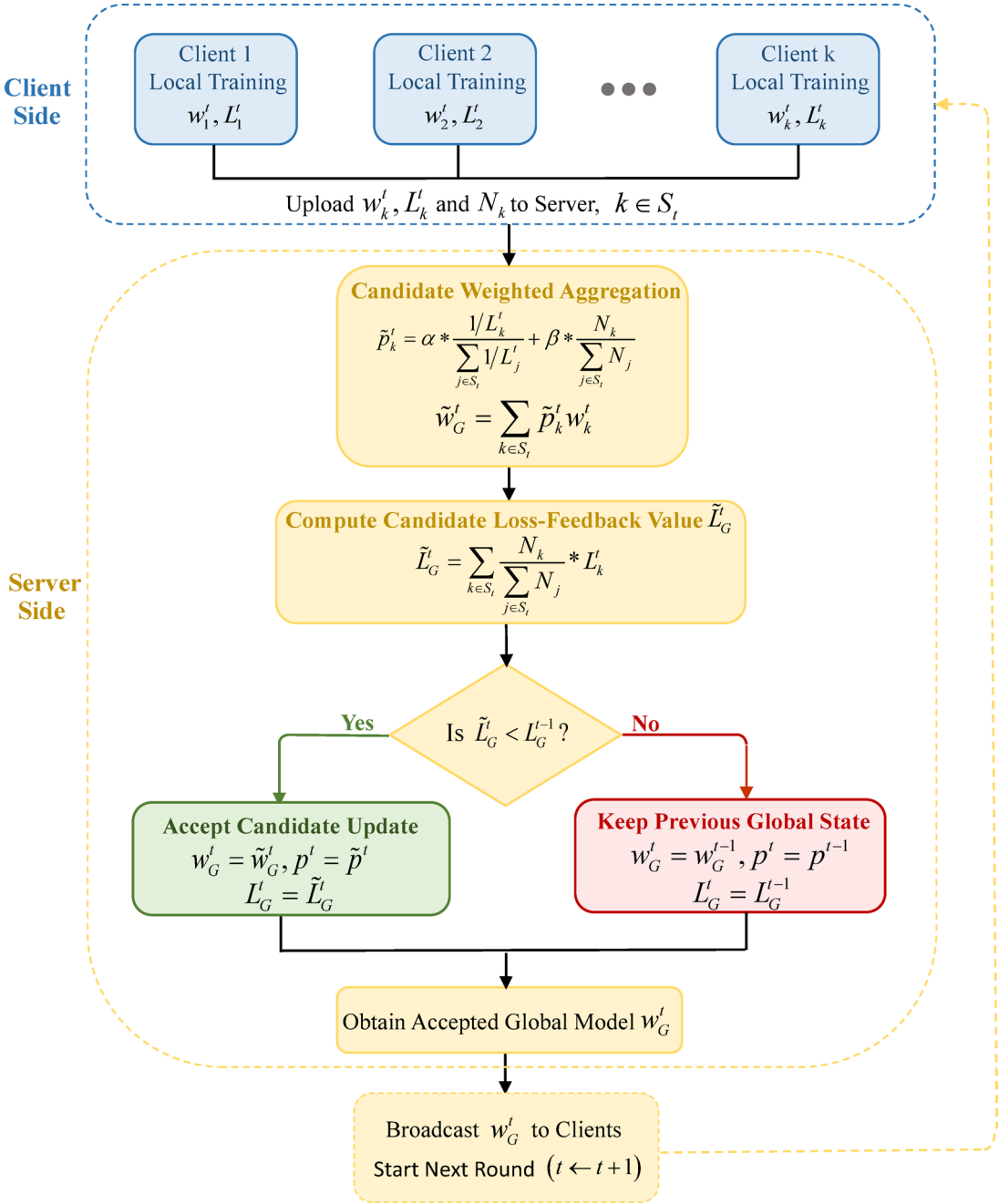


Fig. 3 Flowchart of the loss-feedback-based dynamic weighted aggregation process.

while characterizing the stability of the aggregation rule and the convergence behavior of the accepted loss sequence.

In each communication round t, the central server distributes the global model to all participating clients. Subsequently, each client optimizes the received global model parameters using its local dataset, and the resulting updated model is regarded as the local model for the current communication round. After local optimization is completed, the updated local models are uploaded to the central server, which generates a new global model by applying the proposed dynamic weighting aggregation strategy to the collected local models. The optimization

objective of the global model is defined in Eq. (4), while the optimization loss of the local model is presented in Eq. (3). Algorithm 1 illustrates the training procedure of the proposed method.

Input: total number of communication rounds as T, number of clients as K, number of local epochs as E, client participation ratio ρ, learning rate as η, hyper-parameter as μ, α, β.

Output: final global model parameters w_G^T

Server executes: Initialize $w_G^0, L^0 = +\infty, p^0$
  for $t = 1, 2, \ldots, T$ **do**
    Select the participating client set S_t according to ρ, and broadcast w_G^{t-1}
    for each client $k \in S_t$ in parallel **do**
      Perform E local epochs by minimizing: $L_k^t = L_{CE} + \mu L_{SC}$
      Return the local model parameters w_k^t, local loss L_k^t, and local sample size N_k to the server
    end for
    $\tilde{p}_k^t = \alpha * \frac{1/L_k^t}{\sum_{j \in S_t} 1/L_j^t} + \beta * \frac{N_k}{\sum_{j \in S_t} N_j}$;//Calculate the candidate aggregation weights
    $\tilde{w}_G^t = \sum_{k \in S_t} \tilde{p}_k^t w_k^t$;//Calculate the candidate global model parameters
    $\tilde{L}_G^t = \sum_{k \in S_t} \frac{N_k}{\sum_{j \in S_t} N_j} L_k^t$;//Calculate the candidate loss-feedback value
    if $\tilde{L}_G^t < L_G^{t-1}$
      $\left(w_G^t, p^t, L_G^t\right) \leftarrow \left(\tilde{w}_G^t, \tilde{p}^t, \tilde{L}_G^t\right)$
    else
      $\left(w_G^t, p^t, L_G^t\right) \leftarrow \left(w_G^{t-1}, p^{t-1}, L_G^{t-1}\right)$
    end if
  end for
return w_G^T

Algorithm 1

Experiments

Datasets

To evaluate the effectiveness of the proposed federated learning framework under heterogeneous data distributions, experiments were conducted on two medical X-ray image datasets: the OS Collected Data osteoporosis knee X-ray dataset and the COVID–Pneumonia–Normal chest X-ray dataset. The two datasets contain different



Fig. 4 Sample images from the three categories of the osteoporosis knee X-ray dataset.

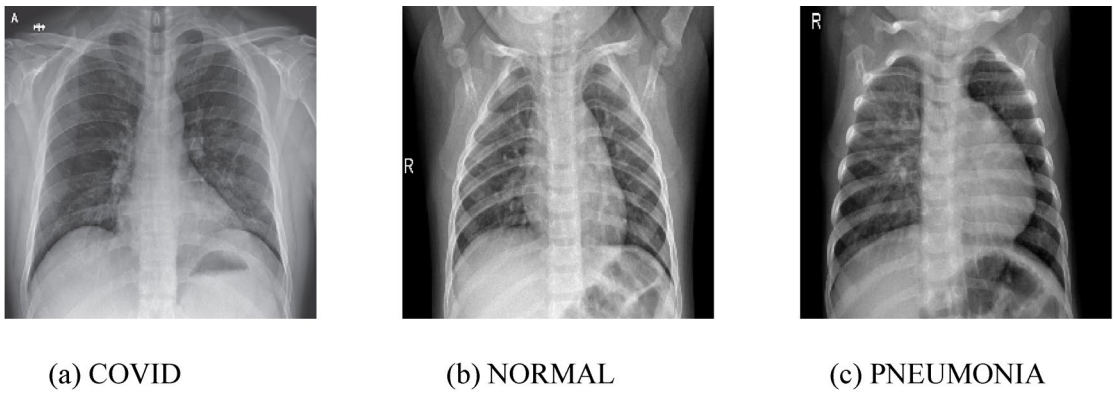


Fig. 5 Sample images from the three categories of the COVID–Pneumonia–Normal chest X-ray dataset.

image categories and data distributions, providing complementary experimental settings for evaluating the proposed method.

The OS Collected Data osteoporosis knee X-ray dataset was constructed by integrating four independent public data sources and contains images acquired using different imaging devices, acquisition parameters, image resolutions, and viewing angles. The dataset contains a total of 1,947 knee X-ray images, which are categorized into three classes according to disease severity: Normal, Osteopenia, and Osteoporosis. Specifically, the Normal, Osteopenia, and Osteoporosis categories contain 780, 374, and 793 images, respectively. Representative samples from the three categories are shown in Fig. 4.

The COVID–Pneumonia–Normal chest X-ray dataset contains a total of 5,228 chest X-ray images and is divided into three categories: COVID, Normal, and Pneumonia. Specifically, the dataset contains 1,626 COVID images, 1,802 Normal images, and 1,800 Pneumonia images. Representative samples from the three categories are shown in Fig. 5.

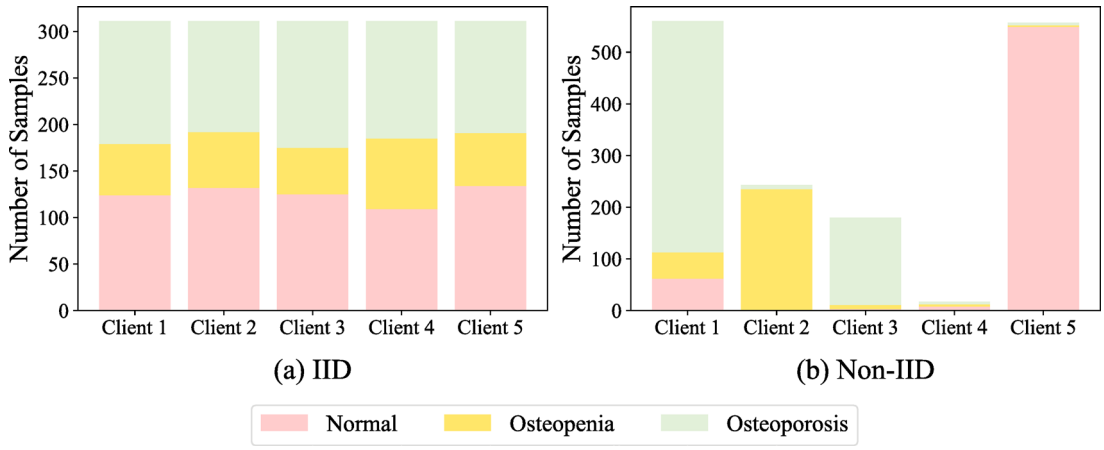


Fig. 6 Distribution of client data categories in the osteoporosis knee X-ray dataset.

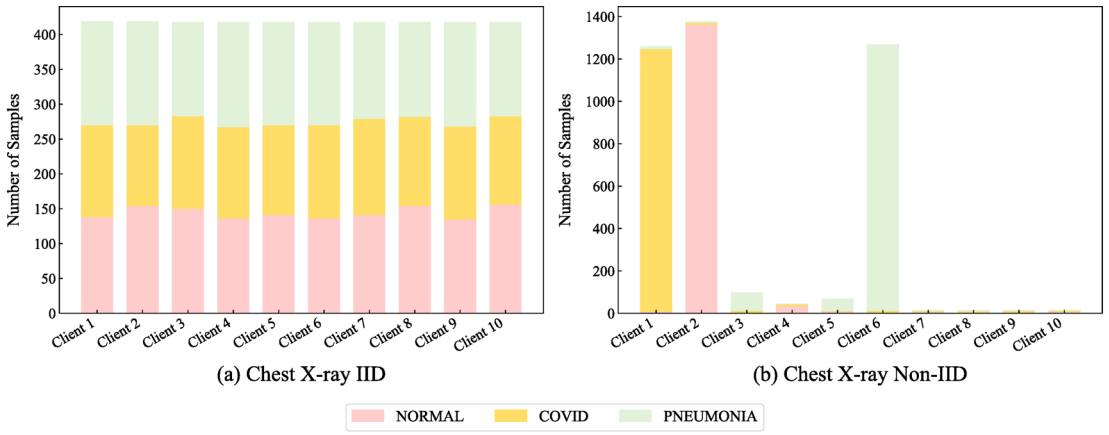


Fig. 7 Distribution of client data categories in the COVID–Pneumonia–Normal chest X-ray dataset.

Experiment scheme

To comprehensively evaluate the performance of the proposed method, both datasets were evaluated under independent and identically distributed (IID) and non-independent and identically distributed (Non-IID) settings. For both datasets, the images were divided into training, validation, and test sets at a ratio of 8:1:1.

For the osteoporosis knee X-ray dataset, the training data were distributed among five clients, and all five clients participated in each communication round, corresponding to a client participation ratio of 1.0. For the COVID–Pneumonia–Normal chest X-ray dataset, the training data were distributed among ten clients. A client participation ratio of 0.8 was adopted, corresponding to eight participating clients in each communication round.

- Independent and identically distributed (IID): The training data are randomly shuffled and evenly allocated among all clients, ensuring that each client maintains a consistent class distribution and thereby establishing an ideal balanced data environment. As illustrated in Fig. 6a and Fig. 7a, each client contains samples from all three classes, and the class distributions across different clients are nearly identical.
- Non-independent and identically distributed (Non-IID): The dataset is partitioned using a Dirichlet distribution, denoted as $Dir(\gamma)$. The parameter γ controls the variation in class distributions across different clients, thereby simulating heterogeneous label distribution scenarios. Specifically, smaller values of γ correspond to

more pronounced discrepancies in class proportions among clients and stronger data heterogeneity, causing individual clients to contain data concentrated in fewer categories.

Figure 6b illustrates the sample distribution under a Dirichlet concentration parameter of $\gamma = 0.1$. The data distributions across different clients exhibit a pronounced label skew, with the number of samples per client ranging from 20 to 560. Specifically, Clients 1 and 3 predominantly contain Osteoporosis samples, Client 2 consists almost entirely of Osteopenia samples, while Client 4 possesses only a limited number of samples with a highly scattered class distribution.

Figure 7b presents the distribution of the 4,182 chest X-ray training images among ten clients under a Dirichlet concentration parameter of $\gamma = 0.1$. The clients exhibit substantial differences in both local dataset size and class composition, with the number of samples per client ranging from 16 to 1,378. Client 1 contains 1,259 samples and is dominated by the COVID category, including 1,242 COVID images. Client 2 contains 1,378 samples and is dominated by the Normal category, including 1,363 Normal images. Client 6 contains 1,269 samples, of which 1,256 belong to the Pneumonia category. Clients 3 and 5 also predominantly contain Pneumonia images, whereas Client 4 mainly contains Normal images. In contrast, Clients 7–10 each contain only 16 samples, with relatively dispersed class compositions.

Overall, the above partitioning scheme provides a challenging Non-IID experimental setting by jointly introducing heterogeneity in client sample sizes and class distributions.

In our experiments, all X-ray images were converted into single-channel grayscale images and resized to 224×224 pixels. The images were normalized using a mean of 0.5 and a standard deviation of 0.5. DenseNet121 was employed as the backbone network, with a batch size of 8, 200 communication rounds, and 3 local training epochs per communication round. The learning rate was set to 2e-4. For local training, we employed the Adam optimizer with exponential decay rates $\beta_1 = 0.9$ and $\beta_2 = 0.99$. The temperature parameter in the supervised contrastive loss was set to 0.5. Each experiment was independently repeated three times, and the results are reported as the mean and standard deviation. Based on the three independent runs, the 95% confidence intervals were further estimated using Student's t-distribution to characterize the uncertainty across runs. We evaluated the classification performance using five commonly used metrics: accuracy, specificity, sensitivity, area under the receiver operating characteristic curve (AUC), and F1 score.

Setting the supervised contrastive loss weight μ

To investigate the influence of the supervised contrastive loss weight, experiments were conducted by assigning different values to the parameter μ in the training loss function defined in Eq. (3). As illustrated in Fig. 8, compared with other settings, $\mu = 1.0$ achieved the highest intermediate accuracy at the 36th communication round and maintained comparatively high performance throughout the subsequent training process. When $\mu = 1.0$, an optimal balance is established between the classification loss and the supervised contrastive loss. This balance enables the model to preserve classification capability while more effectively pulling positive sample representations closer

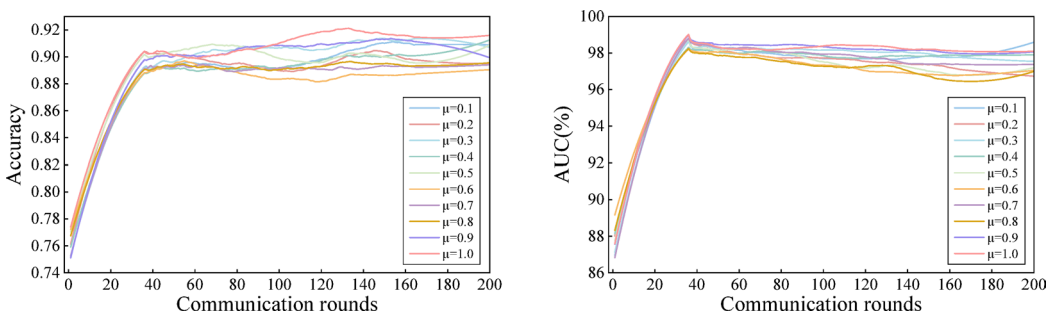


Fig. 8 Effect of different values of μ on model performance.

together and separating them from negative sample representations in the feature space, thereby enhancing the discriminative power of the learned features. Therefore, the supervised contrastive loss weight μ is set to 1.0 in this study.

Configuring the dynamic weighting ratios

To investigate the influence of client-side training loss and local dataset size on model performance within the proposed dynamic weighting aggregation strategy, comparative experiments were conducted on the parameters α and β in Eq. (5). Under the constraint $\alpha + \beta = 1$, six different parameter combinations were evaluated. As illustrated in Fig. 9, all parameter settings exhibited rapid performance improvement during the early training stage, followed by a gradual convergence phase. Among these configurations, the combination $\alpha = 0.4, \beta = 0.6$ achieved the best overall performance, with the final accuracy stabilizing between 0.91 and 0.92 and the AUC remaining around 98%, yielding higher accuracy and AUC values than the other evaluated parameter settings. These results indicate that assigning a relatively larger weight to local dataset size while appropriately considering client training loss can more effectively balance model convergence and aggregation quality. Therefore, the parameter combination $\alpha = 0.4, \beta = 0.6$ was adopted in all subsequent experiments.

The hyperparameter experiments for the supervised contrastive loss weight and dynamic weighting ratios were conducted on the osteoporosis knee X-ray dataset, and the same parameter settings were directly applied to the COVID–Pneumonia–Normal chest X-ray dataset.

Performance comparison with other methods under different levels of data heterogeneity

During the training process, FedAvg, FedProx, and MOON[24] were selected as representative federated learning baselines. FedCL was constructed by retaining the supervised contrastive learning procedure while replacing the proposed dynamic weighted aggregation mechanism with the conventional aggregation strategy. To further isolate the contribution of the aggregation strategy, a static-weighting ablation setting termed FedCL-SW was introduced. FedCL-SW retained the same supervised contrastive learning procedure as FedCL, while its aggregation weights were jointly determined by the local dataset size and local training performance and remained fixed throughout federated training. FedCL-SW was evaluated on both datasets under the IID and severe Non-IID ($\gamma = 0.1$) settings. In the heterogeneity-sweep experiments, FedCL-SW was included only at $\gamma = 0.1$, whereas the other methods were evaluated at $\gamma = 0.1, 0.3, 0.5, 0.7, 0.9$. The corresponding results are presented in Fig. 10, Table 1 and Table 2.

As shown in Fig. 10a, under the highly heterogeneous setting of $\gamma = 0.1$ on the knee X-ray dataset, FedAvg and FedProx exhibit relatively low accuracy and AUC, together with noticeable fluctuations after the initial convergence stage. FedCL and MOON achieve better overall performance, while FedCL-SW further improves the classification results. The proposed method maintains relatively high accuracy and AUC throughout most

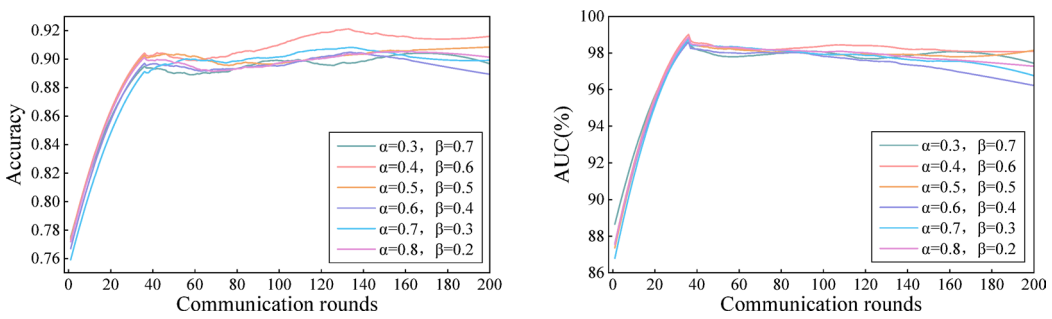


Fig. 9 Effect of different weight ratios on model performance.

https://doi.org/10.1038/s41598-026-72340-1

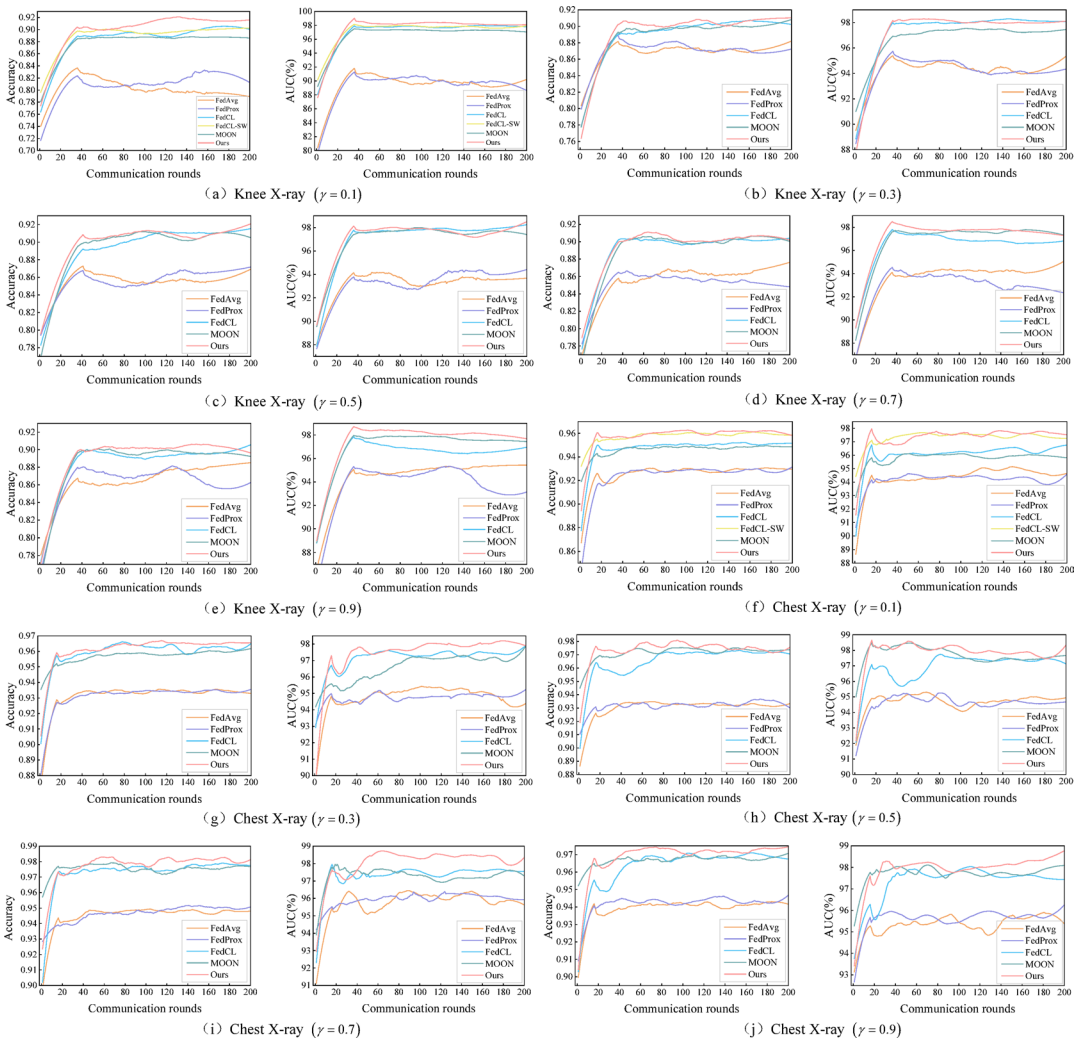


Fig. 10 Performance comparison of different methods under different levels of data heterogeneity.

communication rounds, indicating better convergence performance under severe data heterogeneity. In Fig. 10f, all methods achieve higher and more stable performance on the chest X-ray dataset. FedAvg and FedProx still perform worse than the contrastive learning-based methods, whereas FedCL and MOON show clear improvements. FedCL-SW and the proposed method achieve comparable accuracy, and both outperform the other methods. In addition, the proposed method maintains a relatively high AUC during most of the later communication rounds, demonstrating its effectiveness under highly heterogeneous settings.

As γ increases from 0.1 to 0.9, the degree of data heterogeneity gradually decreases. Under this condition, the performance gap among different methods becomes progressively smaller, and the convergence process exhibits improved stability. Nevertheless, the proposed method continues to maintain comparatively high performance across multiple communication rounds, indicating comparatively stable performance within the evaluated heterogeneity settings. Therefore, in the subsequent experiments, γ is set to 0.1 to further evaluate the effectiveness of the proposed method in highly heterogeneous scenarios.

Table 1 Performance comparison of different methods under various heterogeneous conditions on the knee X-ray dataset.

Dataset		Knee				
Dir(γ)	Methods	Accuracy	Specificity	Sensitivity	AUC	F1
$\gamma = 0.1$	FedAvg	0.8366±0.0266	0.8722±0.0191	0.7955±0.0237	91.8198±0.5085	0.7972±0.0218
	FedProx	0.8329±0.0138	0.8831±0.0255	0.7897±0.0229	89.8930±0.2784	0.7753±0.0134
	FedCL	0.9054±0.0150	0.9369±0.0157	0.9159±0.0122	97.9207±0.1619	0.8753±0.0228
	FedCL-SW	0.9027±0.0162	0.9220±0.0162	0.9122±0.0026	97.9456±0.2958	0.8581±0.0137
	MOON	0.8882±0.0261	0.9116±0.0094	0.9360±0.0153	97.2965±0.1813	0.8698±0.0214
	Ours	0.9212±0.0132	0.9538±0.0068	0.9409±0.0140	98.3117±0.1732	0.9092±0.0229
$\gamma = 0.3$	FedAvg	0.8820±0.0235	0.9211±0.0129	0.8515±0.0068	95.3580±0.2627	0.8420±0.0165
	FedProx	0.8890±0.0114	0.8962±0.0232	0.8872±0.0136	95.7382±0.2743	0.8371±0.0155
	FedCL	0.9076±0.0162	0.9290±0.0104	0.9246±0.0210	98.1948±0.4169	0.8738±0.0010
	MOON	0.9082±0.0266	0.9175±0.0165	0.9255±0.0182	97.4787±0.3739	0.8722±0.0120
	Ours	0.9102±0.0219	0.9453±0.0132	0.9290±0.0171	98.1081±0.6419	0.8957±0.0101
$\gamma = 0.5$	FedAvg	0.8729±0.0181	0.9177±0.0144	0.8522±0.0277	93.8510±0.3232	0.8349±0.0133
	FedProx	0.8718±0.0110	0.9183±0.0176	0.8284±0.0043	94.4203±0.3967	0.8226±0.0095
	FedCL	0.9155±0.0143	0.9418±0.0051	0.9247±0.0155	98.2405±0.6473	0.8856±0.0102
	MOON	0.9121±0.0083	0.9431±0.0119	0.9282±0.0227	97.9988±0.7212	0.8936±0.0137
	Ours	0.9209±0.0105	0.9432±0.0223	0.9332±0.0193	98.5034±0.4679	0.8945±0.0245
$\gamma = 0.7$	FedAvg	0.8763±0.0226	0.9047±0.0212	0.8873±0.0230	95.8125±0.3225	0.8364±0.0167
	FedProx	0.8666±0.0120	0.8857±0.0134	0.8962±0.0173	94.3221±0.5500	0.8276±0.0147
	FedCL	0.9041±0.0033	0.9318±0.0127	0.9332±0.0138	96.8125±0.4425	0.8798±0.0098
	MOON	0.9066±0.0085	0.9381±0.0058	0.9307±0.0154	97.7903±0.1307	0.8794±0.0128
	Ours	0.9112±0.0121	0.9396±0.0043	0.9323±0.0143	97.8353±0.2872	0.8879±0.0083
$\gamma = 0.9$	FedAvg	0.8853±0.0061	0.9163±0.0123	0.8974±0.0150	95.4351±0.3024	0.8437±0.0242
	FedProx	0.8811±0.0272	0.8978±0.0065	0.9168±0.0067	95.3368±0.5732	0.8508±0.0138
	FedCL	0.9057±0.0110	0.9340±0.0103	0.9148±0.0109	96.9589±0.1089	0.8733±0.0106
	MOON	0.9009±0.0157	0.9325±0.0077	0.9163±0.0072	97.9139±0.6750	0.8765±0.0053
	Ours	0.9062±0.0201	0.9383±0.0086	0.9216±0.0113	98.1751±0.5369	0.8825±0.0081

Comparison of model performance in extreme Non-IID and IID scenarios

We compared the proposed method with the representative federated learning baselines FedAvg, FedProx, and MOON, together with the FedCL baseline and the FedCL-SW static-weighting ablation setting, under both IID and Non-IID scenarios ($\gamma = 0.1$). As illustrated in Fig. 11, MOON, FedCL, and FedCL-SW achieved satisfactory performance under both experimental settings. However, compared with these methods, the proposed method generally maintained favorable performance throughout the training process, further validating its effectiveness in heterogeneous federated learning environments.

Tables 3 and 4 present the classification performance of the different methods under the IID and Non-IID settings on the two datasets. Although all methods performed worse under the Non-IID setting than under the IID setting, satisfactory classification results were achieved in both scenarios. On the knee X-ray dataset, under the Non-IID setting, the proposed method improved accuracy and F1 score by approximately 10.60% and 17.27%, respectively, compared with FedProx. It also outperformed FedCL and FedCL-SW in terms of accuracy, sensitivity, and F1 score. On the chest X-ray dataset, under the Non-IID setting, the proposed method improved accuracy, AUC, and F1 score by approximately 1.39%, 1.76%, and 1.33%, respectively, compared with MOON. These results indicate that the proposed method can mitigate the adverse effects of data heterogeneity on the global model and improve its classification performance and training stability.

The comparison among FedCL, FedCL-SW, and the proposed method further evaluates the contribution of the aggregation strategy. On the knee X-ray dataset under the Non-IID setting, FedCL-SW did not consistently outperform FedCL, indicating that fixed multi-factor weighting may be affected by the specific client data distribution. In contrast, the proposed method achieved higher accuracy and F1 scores than both FedCL and FedCL-SW.

Table 2 Performance comparison of different methods under various heterogeneous conditions on the COVID–Pneumonia–Normal chest X-ray dataset.

Dataset		Chest				
Dir(γ)	Methods	Accuracy	Specificity	Sensitivity	AUC	F1
$\gamma = 0.1$	FedAvg	0.9308±0.0230	0.9011±0.0135	0.8965±0.0266	94.6484±0.4845	0.8617±0.0278
	FedProx	0.9315±0.0193	0.9084±0.0090	0.8888±0.0112	94.6105±0.5875	0.8563±0.0025
	FedCL	0.9523±0.0060	0.9574±0.0113	0.9482±0.0098	96.5387±0.2884	0.9003±0.0063
	FedCL-SW	0.9614±0.0008	0.9638±0.0047	0.9425±0.0091	97.6066±0.3242	0.9238±0.0192
	MOON	0.9495±0.0032	0.9676±0.0136	0.9479±0.0062	95.9914±0.4169	0.9200±0.0009
	Ours	0.9627±0.0073	0.9603±0.0215	0.9510±0.0197	97.6767±0.2615	0.9322±0.0110
$\gamma = 0.3$	FedAvg	0.9357±0.0193	0.9040±0.0177	0.9161±0.0298	94.9840±0.4830	0.8813±0.0338
	FedProx	0.9356±0.0097	0.9050±0.0067	0.9021±0.0244	95.2366±0.3678	0.9125±0.0254
	FedCL	0.9662±0.0218	0.9413±0.0201	0.9510±0.0012	97.4944±0.4522	0.9198±0.0133
	MOON	0.9618±0.0016	0.9455±0.0172	0.9471±0.0208	97.8765±0.4183	0.9205±0.0301
	Ours	0.9669±0.0121	0.9536±0.0148	0.9536±0.0161	98.0313±0.5441	0.9268±0.0255
$\gamma = 0.5$	FedAvg	0.9349±0.0133	0.9144±0.0112	0.9038±0.0295	95.3014±0.6939	0.8721±0.0246
	FedProx	0.9367±0.0012	0.9004±0.0098	0.9136±0.0242	94.5610±0.1914	0.8735±0.0150
	FedCL	0.9733±0.0024	0.9621±0.0072	0.9591±0.0101	97.4513±0.0539	0.9128±0.0036
	MOON	0.9754±0.0017	0.9596±0.0220	0.9511±0.0002	97.7719±0.3427	0.9124±0.0042
	Ours	0.9810±0.0038	0.9637±0.0190	0.9620±0.0027	98.3279±0.3828	0.9263±0.0237
$\gamma = 0.7$	FedAvg	0.9494±0.0085	0.9123±0.0081	0.9164±0.0180	96.2092±0.1823	0.8898±0.0307
	FedProx	0.9516±0.0254	0.9115±0.0133	0.9229±0.0154	96.2729±0.4803	0.8990±0.0023
	FedCL	0.9790±0.0031	0.9655±0.0218	0.9484±0.0030	97.6655±0.3054	0.9205±0.0081
	MOON	0.9793±0.0073	0.9633±0.0018	0.9530±0.0332	97.6605±0.3559	0.9181±0.0345
	Ours	0.9830±0.0121	0.9655±0.0232	0.9582±0.0287	98.6185±0.6038	0.9317±0.0362
$\gamma = 0.9$	FedAvg	0.9431±0.0205	0.9327±0.0078	0.9283±0.0029	95.8576±0.2945	0.8975±0.0179
	FedProx	0.9468±0.0085	0.9231±0.0101	0.9130±0.0481	96.2524±0.7461	0.8938±0.0065
	FedCL	0.9710±0.0048	0.9607±0.0027	0.9310±0.0109	97.6289±0.0591	0.9182±0.0159
	MOON	0.9707±0.0024	0.9586±0.0075	0.9503±0.0119	98.0906±0.1028	0.9105±0.0116
	Ours	0.9745±0.0073	0.9625±0.0043	0.9490±0.0159	98.7721±0.6960	0.9278±0.0092

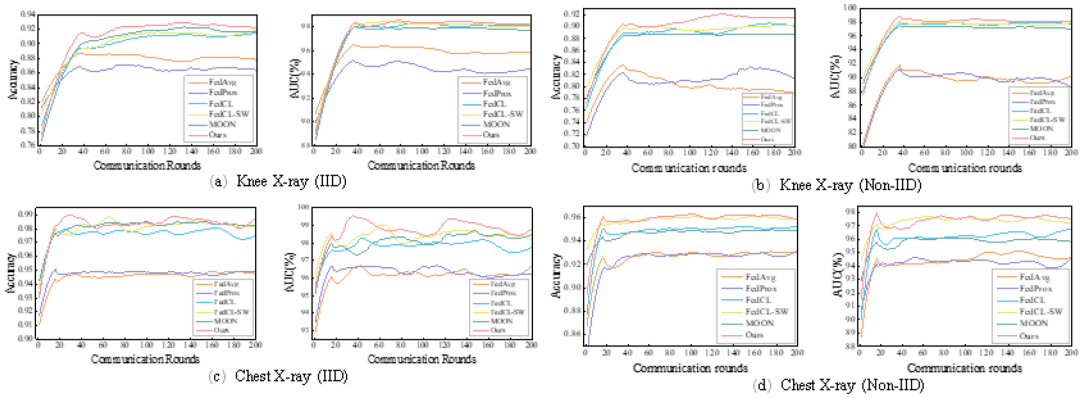


Fig. 11 Performance comparison of different methods under IID and Non-IID ($\gamma = 0.1$) scenarios.

On the chest X-ray dataset, FedCL-SW improved upon FedCL, while the proposed method achieved further improvements in accuracy, sensitivity, AUC, and F1 score. These results indicate that the performance improvement does not arise solely from the introduction of static weighting factors, and that the loss-feedback-based dynamic adjustment contributes to more consistent overall performance across the two datasets.

To further quantify the uncertainty across independent runs, the 95% confidence intervals of accuracy, AUC, and F1 score were calculated under severe Non-IID conditions. The intervals were estimated from three independent runs using Student's t-distribution, and the results are presented in Table 5 as mean [95% confidence

Table 3 Comparison of six federated learning methods on the knee X-ray dataset.

Dataset		Knee				
Schemes	Methods	Accuracy	Specificity	Sensitivity	AUC	F1
IID	FedAvg	0.8876±0.0121	0.9255±0.0276	0.8898±0.0287	96.5254±0.0884	0.8482±0.0135
	FedProx	0.8717±0.0284	0.8833±0.0525	0.8737±0.0056	95.0791±0.1961	0.8407±0.0469
	FedCL	0.9155±0.0066	0.9410±0.0195	0.9282±0.0027	98.0621±0.2659	0.8898±0.0381
	FedCL-SW	0.9192±0.0036	0.9364±0.0197	0.9206±0.0027	98.0221±0.3425	0.8673±0.0370
	MOON	0.9231±0.0028	0.9340±0.0027	0.9197±0.0036	97.9253±0.1771	0.8704±0.0021
	Ours	0.9298±0.0022	0.9572±0.0022	0.9499±0.0003	98.4831±0.1163	0.9144±0.0213
Non-IID	FedAvg	0.8366±0.0266	0.8722±0.0191	0.7955±0.0237	91.8198±0.5085	0.7972±0.0218
	FedProx	0.8329±0.0138	0.8831±0.0255	0.7897±0.0229	89.8930±0.2784	0.7753±0.0134
	FedCL	0.9054±0.0150	0.9369±0.0157	0.9159±0.0122	97.9207±0.1619	0.8753±0.0228
	FedCL-SW	0.9027±0.0162	0.9220±0.0162	0.9122±0.0026	97.9456±0.2958	0.8581±0.0137
	MOON	0.8882±0.0261	0.9116±0.0094	0.9360±0.0153	97.2965±0.1813	0.8698±0.0214
	Ours	0.9212±0.0132	0.9538±0.0068	0.9409±0.0140	98.3117±0.1732	0.9092±0.0229

Table 4 Comparison of six federated learning methods on the COVID–Pneumonia–Normal chest X-ray dataset.

Dataset		Chest				
Schemes	Methods	Accuracy	Specificity	Sensitivity	AUC	F1
IID	FedAvg	0.9490±0.0109	0.9245±0.0068	0.9073±0.0139	96.1929±0.2384	0.8917±0.0031
	FedProx	0.9510±0.0145	0.9371±0.0146	0.9177±0.0024	96.7123±0.3557	0.8939±0.0130
	FedCL	0.9809±0.0073	0.9607±0.0045	0.9376±0.0089	97.9696±0.4984	0.9259±0.0112
	FedCL-SW	0.9887±0.0006	0.9620±0.0030	0.9481±0.0143	97.9839±0.5552	0.9299±0.0072
	MOON	0.9853±0.0047	0.9683±0.0097	0.9535±0.0098	98.5352±0.6258	0.9358±0.0119
	Ours	0.9900±0.0012	0.9770±0.0004	0.9563±0.0129	98.8985±0.3135	0.9480±0.0199
Non-IID	FedAvg	0.9308±0.0230	0.9011±0.0135	0.8965±0.0266	94.6484±0.4845	0.8617±0.0278
	FedProx	0.9315±0.0193	0.9084±0.0090	0.8888±0.0112	94.6105±0.5875	0.8563±0.0025
	FedCL	0.9523±0.0060	0.9574±0.0113	0.9482±0.0098	96.5387±0.2884	0.9003±0.0063
	FedCL-SW	0.9614±0.0008	0.9638±0.0047	0.9425±0.0091	97.6066±0.3242	0.9238±0.0192
	MOON	0.9495±0.0032	0.9676±0.0136	0.9479±0.0062	95.9914±0.4169	0.9200±0.0009
	Ours	0.9627±0.0073	0.9603±0.0215	0.9510±0.0197	97.6767±0.2615	0.9322±0.0110

Table 5 95% confidence intervals of the main performance metrics under severe Non-IID conditions.

Dataset	Methods	Accuracy (95% CI)	AUC (95% CI)	F1 (95% CI)
Knee	FedAvg	0.8366 [0.7705, 0.9027]	91.8198 [90.5566, 93.0830]	0.7972 [0.7430, 0.8514]
	FedProx	0.8329 [0.7986, 0.8672]	89.8930 [89.2014, 90.5846]	0.7753 [0.7420, 0.8086]
	FedCL	0.9054 [0.8681, 0.9427]	97.9207 [97.5185, 98.3229]	0.8753 [0.8187, 0.9319]
	FedCL-SW	0.9027 [0.8625, 0.9429]	97.9456 [97.2108, 98.6804]	0.8581 [0.8241, 0.8921]
	MOON	0.8882 [0.8234, 0.9530]	97.2965 [96.8461, 97.7469]	0.8698 [0.8166, 0.9230]
	Ours	0.9212 [0.8884, 0.9540]	98.3117 [97.8814, 98.7420]	0.9092 [0.8523, 0.9661]
Chest	FedAvg	0.9308 [0.8737, 0.9879]	94.6484 [93.4448, 95.8520]	0.8617 [0.7926, 0.9308]
	FedProx	0.9315 [0.8836, 0.9794]	94.6105 [93.1511, 96.0699]	0.8563 [0.8501, 0.8625]
	FedCL	0.9523 [0.9374, 0.9672]	96.5387 [95.8223, 97.2551]	0.9003 [0.8846, 0.9160]
	FedCL-SW	0.9614 [0.9594, 0.9634]	97.6066 [96.8012, 98.4120]	0.9238 [0.8761, 0.9715]
	MOON	0.9495 [0.9416, 0.9574]	95.9914 [94.9558, 97.0270]	0.9200 [0.9178, 0.9222]
	Ours	0.9627 [0.9446, 0.9808]	97.6767 [97.0271, 98.3263]	0.9322 [0.9049, 0.9595]

interval], with the best mean value for each dataset and metric highlighted in bold. The proposed method achieved the highest mean accuracy, AUC, and F1 score on both datasets. The confidence intervals further characterize the run-to-run uncertainty of the evaluated methods. Although some intervals overlap because only three independent runs were available, the proposed method maintains favorable mean performance across the two datasets.

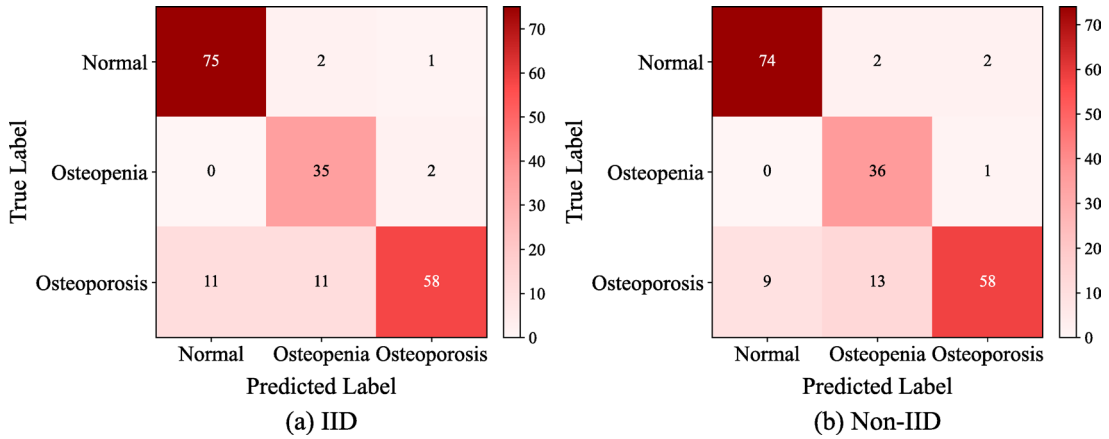


Fig. 12 Confusion matrices of training results under different data distributions on the knee X-ray dataset.

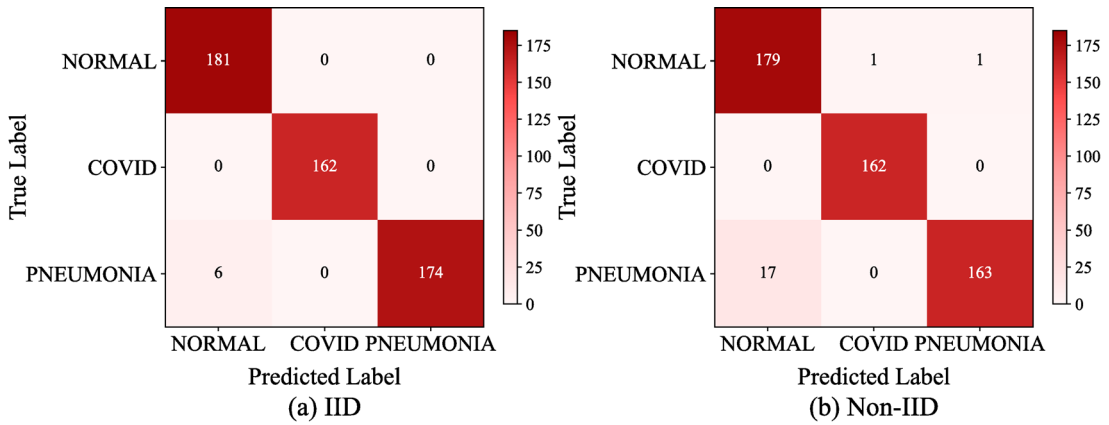


Fig. 13 Confusion matrices of training results under different data distributions on the COVID–Pneumonia–Normal chest X-ray dataset.

As shown in Figs. 12 and 13, the proposed method achieved good classification performance in both scenarios. On the knee X-ray dataset, the recognition rates for the Normal and Osteopenia classes remained above 94% under both IID and Non-IID settings. Under the Non-IID setting on the chest X-ray dataset, the recognition rate for the Normal class reached 98.9%. These results indicate that the proposed method retains good class discrimination under heterogeneous data distributions.

Analysis of the impact of client count on model performance

Subsequently, this paper further investigates the impact of the number of participating clients on model performance in federated learning scenarios. Figure 14 compares the test performance of FedProx, FedCL, and the proposed method on the knee X-ray dataset under both IID and Non-IID settings, with the number of local clients set to 3, 4, 5, 6, 7, and 8. The performance of all three methods generally declines as the number of participating clients increases. This decline is relatively moderate under the IID setting but becomes more pronounced under the Non-IID setting. As the data are distributed across an increasing number of clients, the amount of local data available to each client decreases, while differences among local data distributions exert a stronger influence

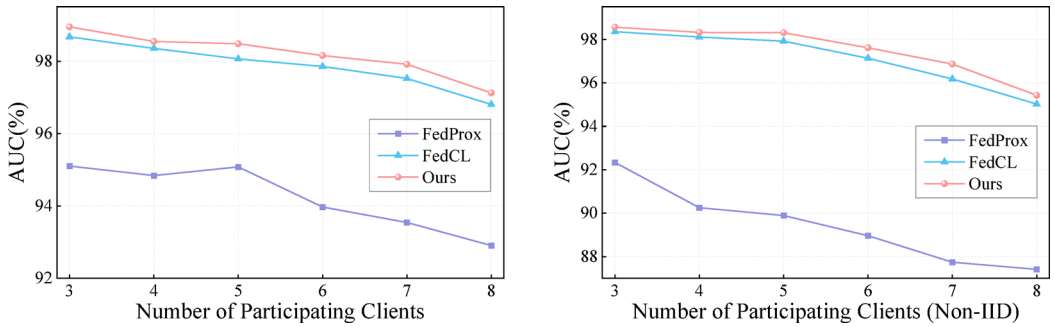


Fig. 14 Classification performance curves for FedProx, FedCL, and the method proposed in this paper as the number of participating clients increases.

on local model training. Across the evaluated client settings, the proposed method consistently achieves favorable overall performance compared with FedProx and FedCL and exhibits a comparatively smaller performance decline.

Table 6 presents the detailed performance results of FedProx, FedCL, and the proposed method with different numbers of participating clients. Under the IID setting, the performance metrics of the three methods generally decline as the number of clients increases, while the AUC of the proposed method remains above 97%. Under the Non-IID setting, the performance degradation becomes more evident because of the combined effects of reduced local data availability and heterogeneous client distributions. Nevertheless, within the evaluated range of 3 to 8 clients, the proposed method maintains comparatively stable classification performance and performs favorably against FedProx and FedCL. These results support the effectiveness of the proposed method under the client configurations examined in this study.

Computational and communication cost analysis

To evaluate the computational cost of the proposed method, the wall-clock time was recorded for complete 200-round training runs under severe Non-IID conditions. On the knee X-ray dataset, the complete training process required approximately 5 h 25 min 12 s, corresponding to an average of 97.56 s per communication round. On the chest X-ray dataset, the complete training process required approximately 10 h 4 min 47 s, corresponding to an average of 181.44 s per communication round. These runtime measurements include client-side local training, server-side aggregation, model evaluation, and logging operations.

The loss-feedback mechanism requires each participating client to upload its local loss and local sample size in addition to the local model parameters in each communication round. Assuming that both values are represented using 32-bit data types and excluding protocol and serialization overhead, the additional algorithmic payload is 40 bytes per round for the five-client knee X-ray setting and 64 bytes per round for the eight-participant chest X-ray setting. Over 200 communication rounds, the corresponding additional payloads are 8,000 bytes and 12,800 bytes, respectively. Apart from these two scalar values, the conventional model-parameter exchange remains unchanged.

Discussion and limitations

The results on two X-ray datasets show that the proposed method maintains stable performance under heterogeneous client distributions. Compared with MOON, the proposed method combines supervised contrastive local training with dynamic weighted aggregation and achieves more consistent results under severe data heterogeneity. FedCL-SW uses the same supervised contrastive learning procedure but adopts static aggregation weights.

Table 6 Performance comparison of FedProx, FedCL, and the proposed method with different numbers of participating clients under IID and Non-IID ($\gamma = 0.1$) conditions.

Schemes	Number of clients	Method	Accuracy	Specificity	Sensitivity	AUC	F1
IID	3	FedProx	0.8873±0.0158	0.9053±0.0133	0.8879±0.0154	95.1027±0.0128	0.8654±0.0184
		FedCL	0.9460±0.0065	0.9672±0.0165	0.9591±0.0247	98.6717±0.0017	0.9302±0.0052
		Ours	0.9562±0.0109	0.9824±0.0057	0.9624±0.0084	98.9486±0.5384	0.9375±0.0031
	4	FedProx	0.8696±0.0146	0.8874±0.0069	0.8622±0.0114	94.8402±0.0095	0.8387±0.0071
		FedCL	0.9372±0.0346	0.9502±0.0181	0.9526±0.0090	98.3541±0.0049	0.9226±0.0043
		Ours	0.9468±0.0145	0.9778±0.0146	0.9563±0.0143	98.5453±0.4557	0.9228±0.0130
	5	FedProx	0.8717±0.0284	0.8833±0.0525	0.8737±0.0056	95.0791±0.1961	0.8407±0.0469
		FedCL	0.9155±0.0066	0.9410±0.0110	0.9282±0.0027	98.0621±0.2659	0.8898±0.0381
		Ours	0.9298±0.0022	0.9572±0.0022	0.9499±0.0003	98.4831±0.1163	0.9144±0.0213
	6	FedProx	0.8650±0.0084	0.8792±0.0062	0.8600±0.0094	93.9712±0.0154	0.8379±0.0060
		FedCL	0.9116±0.0057	0.9385±0.0140	0.9236±0.0121	97.8542±0.0094	0.8823±0.0104
		Ours	0.9168±0.0036	0.9536±0.0063	0.9402±0.0143	98.1555±0.5552	0.9076±0.0072
	7	FedProx	0.8638±0.0394	0.8714±0.0095	0.8429±0.0020	93.5430±0.0085	0.8321±0.0049
		FedCL	0.9248±0.0223	0.9354±0.0020	0.9188±0.0036	97.5241±0.0026	0.8967±0.0052
		Ours	0.9104±0.0097	0.9487±0.0097	0.9352±0.0198	97.9139±0.6258	0.8958±0.0119
	8	FedProx	0.8532±0.0454	0.8606±0.0114	0.8357±0.0026	92.9047±0.0039	0.8329±0.0043
		FedCL	0.9025±0.0133	0.9109±0.0165	0.9048±0.0069	96.8047±0.0175	0.8761±0.0069
		Ours	0.9089±0.0121	0.9373±0.0003	0.9213±0.0082	97.1216±0.3135	0.8848±0.0199
Non-IID	3	FedProx	0.8526±0.0215	0.9035±0.0149	0.8130±0.0043	92.3380±0.0149	0.8061±0.0104
		FedCL	0.9154±0.0161	0.9475±0.0129	0.9226±0.0055	98.3578±0.0094	0.8802±0.0148
		Ours	0.9355±0.0024	0.9782±0.0036	0.9536±0.0086	98.5660±0.3042	0.9265±0.0018
	4	FedProx	0.8399±0.0049	0.8958±0.0188	0.8031±0.0090	90.2520±0.0041	0.7840±0.0030
		FedCL	0.9092±0.0258	0.9413±0.0049	0.9188±0.0104	98.1125±0.0018	0.8782±0.0028
		Ours	0.9236±0.0109	0.9620±0.0072	0.9494±0.0233	98.3253±0.1747	0.9137±0.0153
	5	FedProx	0.8329±0.0138	0.8831±0.0255	0.7897±0.0229	89.8930±0.2784	0.7753±0.0134
		FedCL	0.9054±0.0150	0.9369±0.0157	0.9159±0.0122	97.9207±0.1619	0.8753±0.0228
		Ours	0.9212±0.0132	0.9538±0.0068	0.9409±0.0140	98.3117±0.1732	0.9092±0.0229
	6	FedProx	0.8278±0.0318	0.8603±0.0137	0.7751±0.0216	88.9640±0.0002	0.7708±0.0064
		FedCL	0.9028±0.0145	0.9279±0.0157	0.9123±0.0197	97.1354±0.0057	0.8611±0.0059
		Ours	0.9086±0.0099	0.9414±0.0233	0.9356±0.0058	97.6164±0.6988	0.8827±0.0328
	7	FedProx	0.8316±0.0026	0.8629±0.0077	0.7714±0.0137	87.7430±0.0051	0.7689±0.0036
		FedCL	0.8836±0.0086	0.9215±0.0020	0.9078±0.0020	96.1825±0.0023	0.8554±0.0038
		Ours	0.8956±0.0266	0.9325±0.0264	0.9235±0.0219	96.8670±0.0542	0.8655±0.0431
	8	FedProx	0.8191±0.0123	0.8566±0.0095	0.7409±0.0017	87.4120±0.0015	0.7522±0.0020
		FedCL	0.8857±0.0146	0.9220±0.0134	0.8934±0.0095	95.0252±0.0064	0.8383±0.0033
		Ours	0.8844±0.0048	0.9215±0.0052	0.9136±0.0179	95.4288±0.0345	0.8436±0.0008

Its comparison with the proposed method further supports the effectiveness of dynamic weight adjustment and loss-feedback-based update selection.

However, several limitations remain. The experiments used two public datasets with simulated IID and Dirichlet-based Non-IID partitions, which cannot fully represent distribution shifts caused by different institutions, imaging devices, patient populations, and annotation standards. Local training loss is also used as an indicator of client contribution, although it may be influenced by class imbalance, label noise, and data difficulty. In addition, transmitting model parameters and loss information does not provide a formal privacy guarantee. The current study does not explicitly address security threats such as malicious client updates, model poisoning, or inference attacks. System heterogeneity, including differences in client computing capability, communication bandwidth, and availability, may also affect the practical performance of the proposed method. Moreover, broader comparisons with adaptive aggregation methods and further validation under institutional, regulatory, and network constraints remain necessary. The computational cost and scalability of the method under large-scale, asynchronous, or unstable client participation require further evaluation. Future work will focus on real multi-institutional

validation, more reliable client contribution estimation, privacy-preserving aggregation, and broader medical imaging tasks.

Conclusion

This paper proposes a dynamic weighted federated contrastive learning method for medical image classification under Non-IID data. The method combines supervised contrastive local training with an aggregation strategy based on local training loss and data size, while a loss-feedback rule is used to reject candidate updates that do not reduce the server-side loss-feedback value. Experiments on two X-ray datasets show that the proposed method improves classification performance and maintains greater stability under severe data heterogeneity. Comparisons with representative federated learning methods and the FedCL-SW static-weighting setting further support the effectiveness of the dynamic aggregation mechanism. These results demonstrate the potential of the proposed method for heterogeneous medical image classification, while broader validation in real multi-institutional environments remains necessary.

Author contributions Liu Yang was responsible for the overall study design, development of the main methodology, construction of the experimental framework, data analysis, and manuscript writing and revision. Qinmu Wu provided overall supervision and guidance of the research, contributed to the study design, and was responsible for critical revision and final approval of the manuscript. Jiaheng Li contributed to experimental data analysis and result organization, and assisted in implementing parts of the experiments and model validation. Qingxuan Guan was responsible for initial data preprocessing and organization, and assisted with preliminary experimental work. Qilong Zhao participated in model training and experimental recording, and contributed to summarizing and analyzing experimental results. Boxue Xu contributed to algorithm implementation and code debugging, and supported experiment execution and reproducibility. Ting Yang and Lingfeng Yang assisted in data collection and contributed to manuscript proofreading and review. Tao Zhou contributed to manuscript review and revision.

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Data availability Publicly available in repositories: The datasets analyzed during the current study are available in the following Kaggle repositories. The OS Collected Data osteoporosis knee X-ray dataset is available at https://www.kaggle.com/datasets/mohamedgobara/multi-class-knee-osteoporosis-x-ray-dataset. The COVID–Pneumonia–Normal chest X-ray dataset is available at https://www.kaggle.com/datasets/sachinkumar413/covid-pneumonia-normal-chest-xray-images.

Declarations

Competing interests The authors declare no competing interests.

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References

1. Obuchowicz, R., Strzelecki, M. & Piórkowski, A. Clinical applications of artificial intelligence in medical imaging and image processing—A review. *Cancers* **16**, 1870 (2024).
2. Tsuneki, M. Deep learning models in medical image analysis. *J. Oral Biosci.* **64**, 312–320 (2022).

3. Razzak, M. I., Naz, S. & Zaib, A. Deep learning for medical image processing: Overview, challenges and the future. In *Classification in BioApps: Automation of Decision Making* 323–350 (2017).
4. Appasami, G. & Nickolas, S. A deep learning-based COVID-19 classification from chest X-ray image: Case study. *Eur. Phys. J. Spec. Top.* **231**, 3767–3777 (2022).
5. Appasami, G. & Savarimuthu, N. A novel lightweight CNN design for MRI brain tumor image classification with performance-driven optimization. *Discov. Comput.* **28**, 206 (2025).
6. Mukhtorov, D. et al. Standards-aligned AI validation and certification platform for trustworthy modeling. *IEEE Access* **13**, 216302–216317 (2025).
7. McMahan, B., Moore, E., Ramage, D. et al. Communication-efficient learning of deep networks from decentralized data. *Artif. Intell. Stat.* 1273–1282 (2017).
8. Sheller, M. J. et al. Federated learning in medicine: Facilitating multi-institutional collaborations without sharing patient data. *Sci. Rep.* **10**, 12598 (2020).
9. Olivia, A. Federated deep sense learning with explainability for privacy-preserving diabetes monitoring (2025).
10. Li, W., Milletarì, F., Xu, D. et al. Privacy-preserving federated brain tumour segmentation. *Int. Workshop Mach. Learn. Med. Imaging* 133–141 (2019).
11. Liu, B., Yan, B., Zhou, Y. et al. Experiments of federated learning for COVID-19 chest X-ray images. *arXiv* 2007.05592 (2020).
12. Sana, T. Z. et al. Advancing federated learning: A systematic literature review of methods, challenges, and applications. *IEEE Access* https://doi.org/10.1109/access.2025.3605165 (2025).
13. Wang, X. et al. Securing federated learning with blockchain in the medical field: Systematic literature review. *J. Med. Internet Res.* **28**, e79052 (2026).
14. Li, X. et al. Multi-site fMRI analysis using privacy-preserving federated learning and domain adaptation: ABIDE results. *Med. Image Anal.* **65**, 101765 (2020).
15. Roth, H. R., Chang, K., Singh, P. et al. Federated learning for breast density classification: A real-world implementation. *MICCAI Workshop Domain Adaptation Representation Transfer* 181–191 (2020).
16. Appasami, G. & Savarimuthu, N. Federated learning for secure medical MRI brain tumor image classification. *Eur. Phys. J. Spec. Top.* **234**, 4813–4827 (2025).
17. Appasami, G. & Savarimuthu, N. Cross-dataset performance evaluation and secure federated learning with weighted model aggregation for MRI brain tumor image classification. *J. Supercomput.* **81**, 1555 (2025).
18. Bian, K. & Priyadarshi, R. Machine learning optimization techniques: A survey, classification, challenges, and future research issues. *Arch. Comput. Methods Eng.* **31**, 4209–4233 (2024).
19. Khan, N. et al. Optimizing federated learning with aggregation strategies: A comprehensive survey. *IEEE Open J. Comput. Soc.* https://doi.org/10.1109/ojcs.2025.3590102 (2025).
20. Zhang, C., Xie, Y., Bai, H. et al. A survey on federated learning. *Knowl.-Based Syst.* **216,** 106775 (2021).
21. Li, T. et al. Federated optimization in heterogeneous networks. *Proc. Mach. Learn. Syst.* **2**, 429–450 (2020).
22. Wang, H., Yurochkin, M., Sun, Y. et al. Federated learning with matched averaging. *arXiv* 2002.06440 (2020).
23. Fallah, A., Mokhtari, A. & Ozdaglar, A. Personalized federated learning with theoretical guarantees: A model-agnostic meta-learning approach. *Adv. Neural Inf. Process. Syst.* **33**, 3557–3568 (2020).
24. Chen, X., & He, K. Exploring simple siamese representation learning. In *2021 IEEE/CVF conference on computer vision and pattern recognition (CVPR)* (pp. 15745-15753). IEEE. (2021).
25. Sohan, M. F. & Basalamah, A. A systematic review on federated learning in medical image analysis. *IEEE Access* **11**, 28628–28644 (2023).
26. Sandhu, S. S. et al. Medical imaging applications of federated learning. *Diagnostics* **13**, 3140 (2023).
27. Al-Saleh, A. et al. A federated learning-based privacy-preserving image processing framework for brain tumor detection from CT scans. *Sci. Rep.* **15**, 23578 (2025).
28. Koutsoubis, N. et al. Privacy-preserving federated learning and uncertainty quantification in medical imaging. *Radiol. Artif. Intell.* **7**, e240637 (2025).
29. Nazir, S. & Kaleem, M. Federated learning for medical image analysis with deep neural networks. *Diagnostics* **13**, 1532 (2023).
30. Kaissis, G. A. et al. Secure, privacy-preserving and federated machine learning in medical imaging. *Nat. Mach. Intell.* **2**, 305–311 (2020).
31. Ogier du Terrail, J. et al. Federated learning for predicting histological response to neoadjuvant chemotherapy in triple-negative breast cancer. *Nat. Med.* **29**, 135–146 (2023).
32. Luo, J. & Wu, S. Fedsld: Federated learning with shared label distribution for medical image classification. *IEEE Int. Symp. Biomed. Imaging* 1–5 (2022).
33. He, K., Fan, H., Wu, Y. et al. Momentum contrast for unsupervised visual representation learning. *IEEE/CVF Conf. Comput. Vis. Pattern Recognit.* 9729–9738 (2020).
34. Salloum, R. & Kuo, C. C. J. cPCA++: An efficient method for contrastive feature learning. *Pattern Recognit.* **124**, 108378 (2022).

35. Yin, J. et al. MPCCL: multiview predictive coding with contrastive learning for person re-identification. *Pattern Recognit.* **129**, 108710 (2022).
36. Jing, L. & Tian, Y. Self-supervised visual feature learning with deep neural networks: A survey. *IEEE Trans. Pattern Anal. Mach. Intell.* **43**, 4037–4058 (2020).
37. Misra, I. & Maaten, L. Self-supervised learning of pretext-invariant representations. *IEEE/CVF Conf. Comput. Vis. Pattern Recognit.* 6707–6717 (2020).
38. Yang, X. et al. Heterogeneous graph attention network for unsupervised multiple-target domain adaptation. *IEEE Trans. Pattern Anal. Mach. Intell.* **44**, 1992–2003 (2020).
39. Khosla, P. et al. Supervised contrastive learning. *Adv. Neural Inf. Process. Syst.* **33**, 18661–18673 (2020).
40. Zhang, H., Cao, J., Li, K. et al. Dual-branch contrastive learning for network representation learning. *Int. Conf. Neural Inf. Process.* 185–197 (2023).
41. Yang, H. & Li, J. SimCLIC: A simple framework for contrastive learning of image classification. *J. Syst. Sci. Inf.* **11**, 204–218 (2023).
42. Zhang, Y. et al. Triplet attention and dual-pool contrastive learning for clinic-driven multi-label medical image classification. *Med. Image Anal.* **86**, 102772 (2023).
43. Kuang, Y. et al. PiCCL: A lightweight multiview contrastive learning framework for image classification. *PLoS ONE* **20**, e0329273 (2025).
44. Sohn, K. Improved deep metric learning with multi-class n-pair loss objective. *Adv. Neural Inf. Process. Syst.* **29** (2016).

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